imo-2024-02

An eventually constant greatest common divisor

Arithmétique / Number theory | Level 5/5 (provisional)

Classify the positive integer pairs (a, b) for which the sequence

$$\left(\gcd(a^n + b,\ b^n + a)\right)_{n \geq 1}$$

is eventually constant. Equivalently, positive integers g, N must exist with $\gcd(a^n + b, b^n + a) = g$ whenever the integer n satisfies $n \geq N$.

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Editorial restatement of IMO 2024 problem 2; mathematical conditions preserved, not official wording.

Statement checked against source by Codex. Independent human review pending.